



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO V.T.S. THIRUMURUGAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

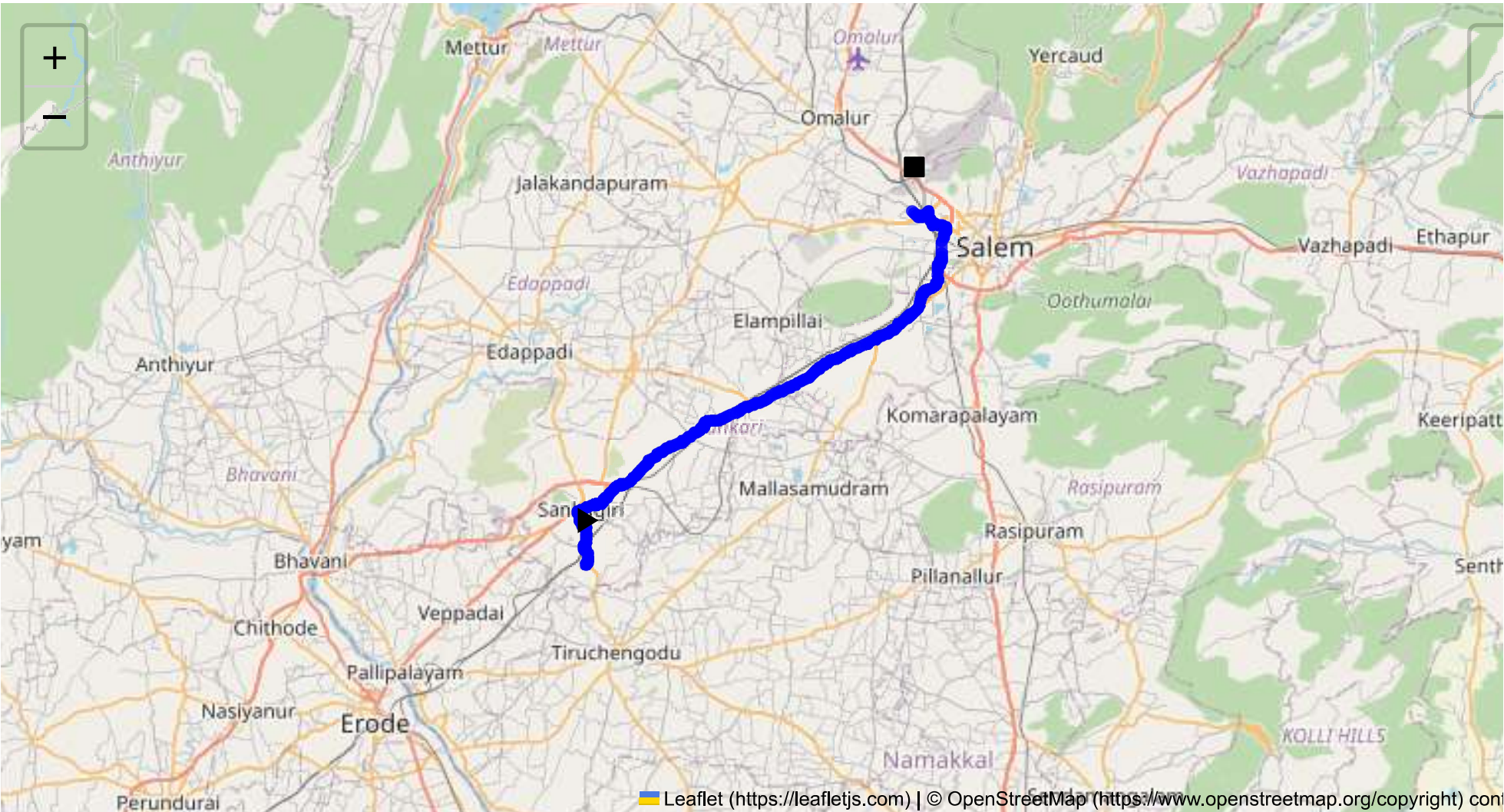
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 45.79 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.678849, 78.100266)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Sundar Nagar in Salem spans approximately 45.79 kilometers in Tamil Nadu, India. It typically takes about 1.03 hours for heavy vehicles, particularly those carrying hazardous materials.

Typical Weather Conditions and Potential Weather-Related Hazards

- **Typical Weather:** Tamil Nadu generally experiences a tropical climate. It includes hot summer months (March to May), a monsoon season (June to November), and a mild winter (December to February).
- **Weather-Related Hazards:** During the monsoon season, heavy rainfall can lead to water-logged roads and reduced visibility. This may cause hydroplaning and increase the risk of accidents. Temperatures can be high in summer months, potentially leading to heat-related issues.

Traffic Patterns Analysis

- **Peak Hours:** Morning (7:00 AM - 10:00 AM) and Evening (5:00 PM - 8:00 PM) on weekdays are typically the peak traffic hours.
- **Congestion-Prone Areas:** Key areas like the NH44 junctions and entry points into Salem city experience frequent traffic congestion.

Road Quality and Infrastructure Assessment

- **Road Quality:** The NH44 highway segment is generally in good condition but be cautious of construction zones or maintenance work. Local roads in Sangagiri and parts of Salem can vary, with potential potholes and uneven surfaces.
- **Infrastructure:** Most intersections have adequate signage, but rural stretches might lack lighting and clear demarcation.

Suggestions for Alternative Routes for Emergencies

An alternative route via NH544 bypass can be considered for emergency purposes. This may add to travel time but provides a safer path away from high-traffic areas.

Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials is subject to specific time restrictions, often barring entry during peak traffic hours.
- Ensure adherence to safe materials handling procedures and compliance with state-guided routes for heavy vehicles.

Overview of Historical Incidents

- There have been occasional incidents involving heavy vehicles, commonly attributed to vehicle breakdowns or improper cargo handling.
- No significant recent hazardous materials incidents have been reported, although general road safety remains an ongoing concern.

Environmental Considerations and Sensitive Areas

- Sensitive areas include water bodies and agricultural land along the route close to the river Kaveri.
- Efforts should be made to minimize spills and emissions, especially when passing through these environmentally sensitive areas.

Communication Coverage

- Most of the route has good mobile network coverage, especially along the NH44. However, some rural patches may have intermittent signal reliability.

Estimated Emergency Response Times

- Urban Areas (Salem): Approximately 15 to 20 minutes.
- Rural Stretches (Sangagiri): Approximately 30 to 45 minutes.

Overall Summary of Risk Assessment

- **Risk Level:** Moderate. The route is generally safe if precautions for weather and traffic-related hazards are followed.
- **Key Risks:** Monsoon season, peak traffic hours, and road quality variances.
- **Recommendations:** Avoid travel during peak hours where possible. Remain vigilant during monsoon rains and ensure vehicle maintenance to mitigate breakdown risks.

Planning and foresight in adhering to regulations, and an understanding of the route dynamics are crucial for managing the transport of hazardous materials on this route safely.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45004, 77.87206	15 KM/Hr
5	Turn	Medium	11.45028, 77.87202	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47410, 77.86845	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.61736, 78.10588	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	Medium	11.61753, 78.10608	30 KM/Hr
12	Turn	Medium	11.64007, 78.11794	30 KM/Hr
13	Turn	Medium	11.64020, 78.11786	30 KM/Hr
14	Turn	Medium	11.64497, 78.12028	30 KM/Hr
15	Turn	Medium	11.64503, 78.12034	30 KM/Hr
16	Turn	High	11.66458, 78.12303	15 KM/Hr
17	Turn	High	11.66477, 78.12298	15 KM/Hr
18	Turn	High	11.66719, 78.12442	15 KM/Hr
19	Turn	Medium	11.66762, 78.12417	30 KM/Hr
20	Turn	Medium	11.67015, 78.11449	30 KM/Hr
21	Turn	Medium	11.67016, 78.11440	30 KM/Hr
22	Turn	Medium	11.67023, 78.11431	30 KM/Hr
23	Turn	Medium	11.67664, 78.11117	30 KM/Hr
24	Turn	Medium	11.67716, 78.11079	30 KM/Hr
25	Turn	High	11.67840, 78.11186	15 KM/Hr
26	Turn	Medium	11.67849, 78.11179	30 KM/Hr
27	Turn	High	11.67851, 78.11172	15 KM/Hr
28	Blind Spot	Blind Spot	11.67472, 78.10494	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
23	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
24	hospital	M.J. Hospital	11.6689574, 78.1209471	30 km/h	Medium
25	police	Suramangalam Police Station	11.6708007, 78.1168863	30 km/h	Medium
26	police	Government Railway Police Station	11.6712204, 78.1130289	30 km/h	Medium
27	hospital	Janani Medical Centre Hospital	11.675896, 78.104826	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



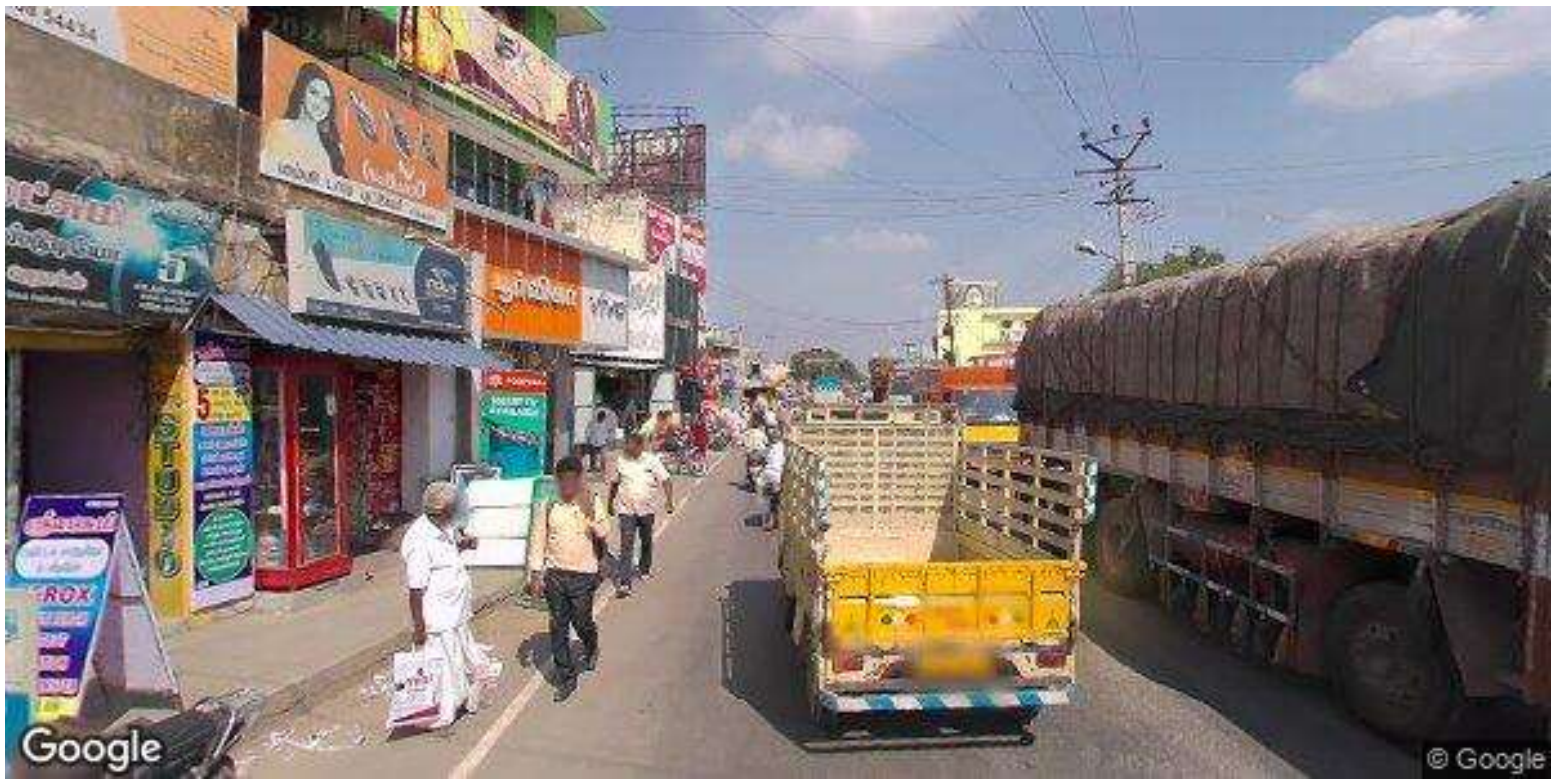
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45004, 77.87206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47410, 77.86845



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61736, 78.10588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61753, 78.10608



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64007, 78.11794



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64020, 78.11786



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64497, 78.12028



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64503, 78.12034



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66458, 78.12303



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66477, 78.12298



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66719, 78.12442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66762, 78.12417



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67015, 78.11449



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67016, 78.11440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67023, 78.11431



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67664, 78.11117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67716, 78.11079



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67840, 78.11186



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67849, 78.11179



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67851, 78.11172



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.67472, 78.10494

